

Pleiotropy Answers:

- 1) Which of the SNPs have genome-wide significant ($p < 5 \times 10^{-8}$) associations for both traits?

rs138, rs139, rs140, rs141, rs296, rs299, rs1138, rs1448

- 2) Did the multivariate analyses result in additional SNPs that had genome-wide significant cross phenotype associations but that also had genome-wide suggestive ($p < 5 \times 10^{-6}$) univariate association for each trait? Which SNP(s)?

rs125, rs135, rs137, rs295, rs298, rs300, rs920, rs921, rs923, rs1166, rs1361

Instead of running mediation analysis on all 19 SNPs, I suggested that you perform LD clumping to reduce this number of SNPs and only focus on the index SNP for each clump (or if the index SNP was not associated with both traits to choose another SNP from among the clumped SNPs). This reduced the set of SNPs to 11.

- 3) For each SNP analyzed in the mediation analysis, determine if there is a significant direct effect which is indicative of some level of biological pleiotropy. Do any of the SNPs exhibit complete mediation?

All SNPs show a significant direct effect on Trait 2 indicating some level of biological pleiotropy. rs923 has an ADE p-value of 0.002 but this is still less than the Bonferroni corrected p-value of 0.0045, adjusting for the 11 SNPs. No SNP shows an association with Trait 2 that is completely mediated through its association with Trait 1, i.e. an ACME estimate that is equal to (or close to) the total effect. The strongest mediated effect is for rs921 in which the mediated effect accounts for ~40% of the total effect of the SNP on Trait 2.

- 4) Why do some of the SNPs have negative values for the proportion mediated?

The estimate of the proportion mediated is not the best way to interpret the mediation results, despite its seemingly obvious interpretability. In reality this proportion does not range from 0-1 but can rather be less than 0 and greater than 1. The negative proportion mediated values that we see for many of the SNPs we have analyzed is due to the fact that these SNPs have an effect estimate for the total effect and mediated effect that are in opposite directions, i.e. the effect of the SNPs on Trait 1 and Trait 2 is in opposite directions. Depending on your study question, you may want to limit your selection of SNPs to only those with effects on the two traits in the same direction. We did not see this among our SNPs, but a proportion mediated > 1 can happen when the strength of the association with the mediator (Trait 1) is much higher than the strength of the association with the outcome (Trait 2). This is often why it is recommended that the direct, indirect and total effects be used in the interpretation rather than the proportion mediated.

Summary table of pleiotropy results

SNP	Beta (Trait 1)	P (Trait 1)	Beta (Trait 2)	P (Trait 2)	MV ¹ P	MV ¹ Beta (Trait 1)	MV ¹ Beta (Trait 2)	ADE	ADE (P)	ACME	ACME (P)	Total Effect	Total Effect (P)	Prop Mediated	Prop Mediated (P)
rs125	0.072	1.08E-08	0.062	6.45E-07	1.80E-10	0.8397	0.7096	0.045	<2e-16	0.015	<2e-16	0.0596	<2e-16	0.2516	<2e-16
rs135	-0.040	1.41E-06	0.050	1.47E-09	2.82E-17	-0.5457	0.7024	0.059	<2e-16	-0.008	<2e-16	0.0509	<2e-16	-0.3317	<2e-16
rs139	-0.065	2.77E-15	0.090	2.86E-28	2.26E-50	-0.5249	0.7196	0.103	<2e-16	-0.014	<2e-16	0.0891	<2e-16	-0.1580	<2e-16
rs296	-0.056	1.15E-10	-0.051	2.60E-09	1.78E-14	0.8100	0.7456	-0.039	<2e-16	-0.012	<2e-16	-0.0512	<2e-16	0.2306	<2e-16
rs300	-0.046	1.85E-08	-0.039	2.09E-06	1.34E-10	-0.8386	-0.7110	-0.029	<2e-16	-0.010	<2e-16	-0.0392	<2e-16	0.2507	<2e-16
rs921	0.109	6.29E-23	0.057	1.95E-07	1.16E-24	-0.9475	-0.5144	0.036	<2e-16	0.023	<2e-16	0.0595	<2e-16	0.3908	<2e-16
rs923	0.041	1.48E-06	0.042	4.04E-07	2.35E-09	0.7615	0.7957	0.034	0.002	0.009	<2e-16	0.0421	<2e-16	0.2035	<2e-16
rs1138	-0.050	9.58E-10	0.048	2.90E-09	2.44E-20	0.6511	-0.6027	0.058	<2e-16	-0.011	<2e-16	0.0468	<2e-16	-0.2319	<2e-16
rs1166	-0.051	4.77E-10	0.041	6.30E-07	1.02E-17	-0.7175	0.5275	0.049	<2e-16	-0.011	<2e-16	0.0382	<2e-16	-0.2918	<2e-16
rs1361	-0.056	2.65E-07	0.076	1.68E-12	3.52E-21	-0.5349	0.7114	0.087	<2e-16	-0.012	<2e-16	0.0751	<2e-16	-0.1614	<2e-16
rs1448	-0.079	3.46E-08	0.092	1.51E-11	2.21E-20	-0.5847	0.6679	0.108	<2e-16	-0.017	<2e-16	0.0908	<2e-16	-0.1879	<2e-16

¹Multivariate